

For More Visit **VU Answer**

MTH301 Final Term Papers By Waqar (File 4)

Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

Which one of the following is correct Wallis Cosine formula when n is odd and $n \geq 3$?

Answer (Please select your correct option)

VuAnswers.com

- ☐ $\int_0^{\frac{\pi}{2}} \cos^n x \, dx = \frac{\pi}{2} \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} \dots \frac{5}{6} \frac{3}{4} \frac{1}{2}$
- ☐ $\int_0^{\frac{\pi}{2}} \cos^n x \, dx = \frac{\pi}{2} \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} \dots \frac{6}{5} \frac{4}{3} \frac{2}{1}$
- ☐ $\int_0^{\frac{\pi}{2}} \cos^n x \, dx = \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} \dots \frac{6}{5} \frac{4}{3} \frac{2}{1}$
- ☐ $\int_0^{\frac{\pi}{2}} \cos^n x \, dx = \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} \dots \frac{6}{7} \frac{4}{5} \frac{2}{3}$

correct Made by: Waqar Siddhu

Question No : 2 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the amplitude of a periodic function defined by $f(x) = 4 \cos 3x$?

Answer (Please select your correct option)

VuAnswers.com

- ☐ 1
- ☐ 3
- ☐ 4
- ☐ 12

correct

Made by: Waqar Siddhu

Question No : 3 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the period of a periodic function defined by $f(x) = \sin \frac{x}{2}$?

Answer (Please select your correct option)

VuAnswers.com

- ☐ $\frac{\pi}{2}$
- ☐ π
- ☐ $\frac{3\pi}{2}$
- ☐ 4π

correct

Made by: Waqar Siddhu

For More Visit **VU Answer**

For More Visit **VU Answer**

Question No : 4 of 52

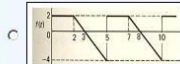
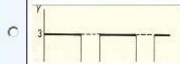
Marks: 1 (Budgeted Time 1 Min)

Match the following periodic function with its graph.

$$f(x) = \begin{cases} 4 & 0 < x < 5 \\ 0 & 5 < x < 8 \end{cases}$$

Answer (Please select your correct option)

VuAnswers.com



Made by: Waqar Siddhu

Question No : 4 of 52

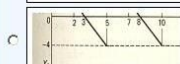
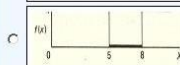
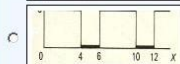
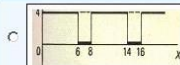
Marks: 1 (Budgeted Time 1 Min)

Match the following periodic function with its graph.

$$f(x) = \begin{cases} 4 & 0 < x < 5 \\ 0 & 5 < x < 8 \end{cases}$$

Answer (Please select your correct option)

VuAnswers.com



Made by: Waqar Siddhu

Question No : 5 of 52

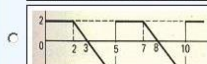
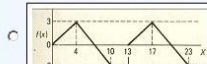
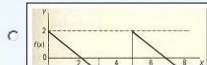
Marks: 1 (Budgeted Time 1 Min)

Match the following periodic function with its graph.

$$f(x) = \begin{cases} 2-x & 0 < x < 3 \\ -1 & 3 < x < 5 \end{cases}$$

Answer (Please select your correct option)

VuAnswers.com



Made by: Waqar Siddhu

For More Visit **VU Answer**

For More Visit **VU Answer**

Question No : 5 of 52

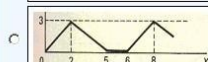
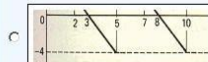
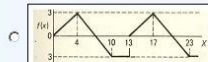
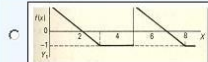
Marks: 1 (Budgeted Time 1 Min)

Match the following periodic function with its graph.

$$f(x) = \begin{cases} 2-x & 0 < x < 3 \\ -1 & 3 < x < 5 \end{cases}$$

Answer (Please select your correct option)

VuAnswers.com



Made by: Waqar Siddhu

Question No : 6 of 52

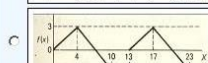
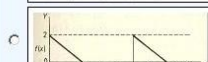
Marks: 1 (Budgeted Time 1 Min)

Match the following periodic function with its graph.

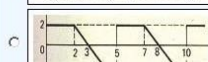
$$f(x) = \begin{cases} \frac{3}{4}x & 0 < x < 4 \\ 7-x & 4 < x < 10 \\ -3 & 10 < x < 13 \end{cases}$$

Answer (Please select your correct option)

VuAnswers.com



correct



Made by: Waqar Siddhu

Question No : 6 of 52

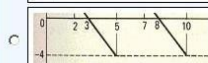
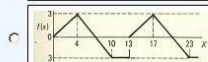
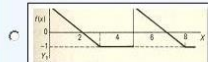
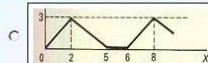
Marks: 1 (Budgeted Time 1 Min)

Match the following periodic function with its graph.

$$f(x) = \begin{cases} \frac{3}{4}x & 0 < x < 4 \\ 7-x & 4 < x < 10 \\ -3 & 10 < x < 13 \end{cases}$$

Answer (Please select your correct option)

VuAnswers.com



Made by: Waqar Siddhu

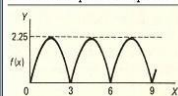
For More Visit **VU Answer**

For More Visit **VU Answer**

Question No : 7 of 52

Marks: 1 (Budgeted Time 1 Min)

What is the period of periodic function whose graph is as below?



Answer (Please select your correct option)

VuAnswers.com

☐

0

☐

2.25

☒

3

correct

☐

6

Made by: Waqar Siddhu

Question No : 8 of 52

Marks: 1 (Budgeted Time 1 Min)

The function $f(x) = x^2 e^x$ is -----

Answer (Please select your correct option)

VuAnswers.com

☐

Even function

☐

Odd function

☒

Neither even nor odd

correct

Made by: Waqar Siddhu

Question No : 9 of 52

Marks: 1 (Budgeted Time 1 Min)

The graph of an even function is symmetrical about -----

Answer (Please select your correct option)

VuAnswers.com

☐

x-axis

☒

y-axis

correct

☐

origin

Made by: Waqar Siddhu

For More Visit **VU Answer**

For More Visit **VU Answer**

Question No : 10 of 52

Marks: 1 (Budgeted Time 1 Min)

Which of the following is Laplace inverse transform of the function $f(s) = \frac{3}{s-2} - \frac{2}{s}$?

☐ $3te^{2t} - 2$

☐ $3e^{2t} - 2t$

☐ $3e^{2t} - 2$

☐ None of these.

correct

VuAnswers.com

Made by: Waqar Siddhu

Question No : 11 of 52

Marks: 1 (Budgeted Time 1 Min)

What is the value of $L(e^{5t})$ if L denotes laplace transform?

☐ $L(e^{5t}) = \frac{1}{s-5}$

☐ $L(e^{5t}) = \frac{s}{s^2+25}$

☐ $L(e^{5t}) = \frac{5}{s^2+25}$

☐ $L(e^{5t}) = \frac{5!}{s^6}$

correct

VuAnswers.com

Made by: Waqar Siddhu

Question No : 12 of 52

Marks: 1 (Budgeted Time 1 Min)

What is laplace transform of the function $F(t)$ if $F(t) = \sin 3t$?

☐ $L(\sin 3t) = \frac{3}{s^2+9}$

☐ $L(\sin 3t) = \frac{s}{s^2+9}$

☐ $L(\sin 3t) = \frac{1}{s-3}$

☐ $L(\sin 3t) = \frac{3!}{s^4}$

correct

VuAnswers.com

Made by: Waqar Siddhu

For More Visit **VU Answer**

For More Visit **VU Answer**

Question No : 13 of 52

Marks: 1 (Budgeted Time 1 Min)

If L denotes laplace transform then
 $L(te^{5t}) =$

☐ $L(te^{5t}) = \frac{1}{s^2 - 5}$

☐ $L(te^{5t}) = \frac{1}{s^2 + 5}$

☐ $L(te^{5t}) = \frac{1}{(s+5)^2}$

☐ $L(te^{5t}) = \frac{1}{(s-5)^2}$

correct

Made by: Waqar Siddhu

Question No : 14 of 52

Marks: 1 (Budgeted Time 1 Min)

What is Laplace Inverse Transform of $\frac{s}{s^2 + 25}$

☐ $L^{-1}\left\{\frac{s}{s^2 + 25}\right\} = \sin 5t$

☐ $L^{-1}\left\{\frac{s}{s^2 + 25}\right\} = \cos 5t$

☐ $L^{-1}\left\{\frac{s}{s^2 + 25}\right\} = \sin 25t$

☐ $L^{-1}\left\{\frac{s}{s^2 + 25}\right\} = \cos 25t$

Made by: Waqar Siddhu

Question No : 15 of 52

Marks: 1 (Budgeted Time 1 Min)

Divergence of a vector function is always a -----

☐ Scalar

correct

☐ Vector

Made by: Waqar Siddhu

For More Visit **VU Answer**

For More Visit **VU Answer**

Question No : 16 of 52

Marks: 1 (Budgeted Time 1 Min)

If p is the period of a function then that function is said to be periodic if $f(x+p) = f(x)$, _____

Answer (Please select your correct option)

VuAnswers.com

☐ For all values of x in the domain of f

correct

☐ For positive values of x in the domain of f

☐ For negative values of x in the domain of f

Made by: Waqar Siddhu

Question No : 17 of 52

Marks: 1 (Budgeted Time 1 Min)

Which of the following is associated to each point on a plane?

Answer (Please select your correct option)

VuAnswers.com

☐ A real number

correct

☐ A natural number

☐ An ordered pair

Made by: Waqar Siddhu

Question No : 18 of 52

Marks: 1 (Budgeted Time 1 Min)

Consider the interval $(-1, 5)$ on co-ordinate line. What does this interval denote?

Answer (Please select your correct option)

VuAnswers.com

☐ The set of all real numbers between -1 and 5

correct

☐ The set of all integers between -1 and 5

☐ The set of all natural numbers between -1 and 5

☐ The set of all rational numbers between -1 and 5

Made by: Waqar Siddhu

For More Visit **VU Answer**

For More Visit **VU Answer**

Question No : 19 of 52

Marks: 1 (Budgeted Time 1 Min)

Equation of the circular disk with radius a and origin at $(0,0)$ is given by

Answer (Please select your correct option)

VuAnswers.com

☐ $x^2 + y^2 = a^2$

correct

☐ $x^2 + y^2 \geq a^2$

☐ $x^2 + y^2 \leq a^2$

Made by: Waqar Siddhu

Question No : 20 of 52

Marks: 1 (Budgeted Time 1 Min)

In three dimensional space, the equation $y = x^2$ always represents

Answer (Please select your correct option)

VuAnswers.com

☐ Parabola

correct

☐ Straight line

☐ Half cylinder

☐ Cone

Made by: Waqar Siddhu

Question No : 21 of 52

Marks: 1 (Budgeted Time 1 Min)

Domain of the function $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$ is

Answer (Please select your correct option)

VuAnswers.com

☐ Entire 3D-Space

correct

☐ Entire 3D-Space except origin

☐ First octant of 3D-Space

Made by: Waqar Siddhu

For More Visit **VU Answer**

For More Visit **VU Answer**

Question No : 22 of 52

Marks: 1 (Budgeted Time 1 Min)

Suppose $f(x, y) = x^3 e^{xy}$. Which of the following options is correct?

Answer (Please select your correct option)

VuAnswers.com

☐ $\frac{\partial f}{\partial y} = 3x^3 e^{xy}$

☐ $\frac{\partial f}{\partial y} = x^3 e^{xy}$

☐ $\frac{\partial f}{\partial y} = x^4 e^{xy}$

☐ $\frac{\partial f}{\partial y} = x^3 y e^{xy}$

correct

Made by: Waqar Siddhu

Question No : 23 of 52

Marks: 1 (Budgeted Time 1 Min)

Gradient of a scalar function always results in a function.

Answer (Please select your correct option)

VuAnswers.com

☐ Scalar

☐ Continuous

☐ Vector

☐ Constant

correct

Made by: Waqar Siddhu

Question No : 24 of 52

Marks: 1 (Budgeted Time 1 Min)

Let x, y, z be the length, width and height of a rectangular box. The area of bottom will be

Answer (Please select your correct option)

VuAnswers.com

☐ $A = yz$

☐ $A = xz$

☐ $A = xy$

☐ $A = xyz$

correct

Made by: Waqar Siddhu

For More Visit **VU Answer**

For More Visit **VU Answer**

Question No : 25 of 52

Marks: 1 (Budgeted Time 1 Min)

Let x, y, z be the length, width and height of an open rectangular box. The surface area of the box will be

Answer (Please select your correct option)

VuAnswers.com

☐ $A = xy + 2yz + 2xz$

☐ $A = yz + 4$

☐ $A = xz + yz + zx$

☐ $A = xyz$

correct

Made by: Waqar Siddhu

Question No : 26 of 52

Marks: 1 (Budgeted Time 1 Min)

If $R = \{(x, y) : 0 \leq x \leq 2 \text{ and } 0 \leq y \leq 3\}$, then $\iint_R (1 - ye^{xy}) dA = \dots\dots\dots$

Answer (Please select your correct option)

VuAnswers.com

☐ $\int_0^2 \int_0^3 (1 - ye^{xy}) dy dx$

correct

☐ $\int_0^2 \int_0^3 (1 - ye^{xy}) dx dy$

☐ $\int_2^3 \int_0^0 (1 - ye^{xy}) dx dy$

☐ $\int_0^2 \int_0^3 (4xe^{2y}) dy dx$

Made by: Waqar Siddhu

Question No : 27 of 52

Marks: 1 (Budgeted Time 1 Min)

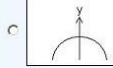
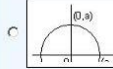
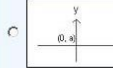
Match the following equation in polar co-ordinates with its graph.

$r \sin \theta = a$

where a is an arbitrary constant

Answer (Please select your correct option)

VuAnswers.com



Made by: Waqar Siddhu

For More Visit **VU Answer**

For More Visit **VU Answer**

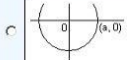
Question No : 27 of 52

Marks: 1 (Budgeted Time 1 Min)

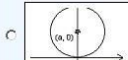
Match the following equation in polar co-ordinates with its graph.
 $r \sin \theta = a$
where a is an arbitrary constant

Answer (Please select your correct option)

VuAnswers.com



correct



Made by: Waqar Siddhu

Question No : 28 of 52

Marks: 1 (Budgeted Time 1 Min)

Polar co-ordinates of a point are $(-8, 45^\circ)$. Which of the following is another possible polar co-ordinates representation of this point?

Answer (Please select your correct option)

VuAnswers.com

☐ $(8, 225^\circ)$

☐ $(-8, 225^\circ)$

☐ $(8, 45^\circ)$

☐ $(8, 315^\circ)$

correct

Made by: Waqar Siddhu

Question No : 29 of 52

Marks: 1 (Budgeted Time 1 Min)

Polar co-ordinates of a point are $\left(7, \frac{-\pi}{4}\right)$. Which of the following is another possible polar co-ordinates representation of this point?

Answer (Please select your correct option)

VuAnswers.com

☐ $\left(7, \frac{3\pi}{4}\right)$

☐ $\left(-7, \frac{3\pi}{4}\right)$

☐ $\left(-7, \frac{-\pi}{4}\right)$

☐ $\left(7, \frac{-3\pi}{4}\right)$

Made by: Waqar Siddhu

For More Visit **VU Answer**

For More Visit **VU Answer**

Question No : 30 of 52

Marks: 1 (Budgeted Time 1 Min)

If $p(r, \theta)$ is a point in polar coordinate system, then r is the distance of p from

Answer (Please select your correct option)

VuAnswers.com

☐ Pole

correct

☐ Imaginary axis

☐ None of these

☐ Polar axis

Made by: Waqar Siddhu

Question No : 31 of 52

Marks: 1 (Budgeted Time 1 Min)

Given the integral $\iint_R f(x, y) dx dy$, after converting to polar coordinates, it will become where $a \leq \theta \leq b$ and $c \leq r \leq d$.

Answer (Please select your correct option)

VuAnswers.com

☐ $\int_a^b \int_c^d f(r, \theta) r dr d\theta$

correct

☐ $\int_a^b \int_c^d f(r, \theta) dr d\theta$

☐ $\int_a^b \int_c^d f(r, \theta) r d\theta dr$

☐ $\int_a^b \int_c^d f(r, \theta) d\theta dr$

Made by: Waqar Siddhu

Question No : 32 of 52

Marks: 1 (Budgeted Time 1 Min)

The graph of the equation $r = 3\hat{i} - 2\hat{j} - \hat{k}$ is the point

Answer (Please select your correct option)

VuAnswers.com

☐ (1, -2, 3)

☐ (3, -2, -1)

correct

☐ (3, 2, 1)

☐ (-3, -2, -1)

Made by: Waqar Siddhu

For More Visit **VU Answer**

For More Visit **VU Answer**

Question No : 33 of 52

Marks: 1 (Budgeted Time 1 Min)

The natural domain of a vector valued function is the of the natural domains of its components

Answer (Please select your correct option)

VuAnswers.com

☐ Product

☐ Summation

☐ Union

☐ Intersection

correct

Made by: Waqar Siddhu

Question No : 34 of 52

Marks: 1 (Budgeted Time 1 Min)

If $\vec{R}(t)$ is the anti-derivative of a given vector valued function $\vec{r}(t)$ i.e., $\vec{R}'(t) = \vec{r}(t)$ then $\int_a^b \vec{r}(t) dt = \dots\dots\dots$

Answer (Please select your correct option)

VuAnswers.com

☐ $\vec{R}(a) + \vec{R}(b)$

☐ $\vec{R}(b) - \vec{R}(a) + c$ where c is a non-zero constant.

☐ $\vec{R}(a) \cdot \vec{R}(b)$

☐ $\vec{R}(b) - \vec{R}(a)$

correct

Made by: Waqar Siddhu

Question No : 35 of 52

Marks: 1 (Budgeted Time 1 Min)

For the given vector valued functions $\vec{r}_1(t)$ and $\vec{r}_2(t)$, $\frac{d}{dt}[\vec{r}_1(t) \times \vec{r}_2(t)] = \dots\dots\dots$ where " $\times$ " and " $\cdot$ " represent the cross and dot product respectively.

Answer (Please select your correct option)

VuAnswers.com

☐ $\vec{r}_1 \cdot \frac{d\vec{r}_2}{dt} + \frac{d\vec{r}_1}{dt} \cdot \vec{r}_2$

☐ $\vec{r}_1 \times \frac{d\vec{r}_2}{dt} + \frac{d\vec{r}_1}{dt} \times \vec{r}_2$

☐ $\frac{d\vec{r}_2}{dt} \times \vec{r}_1 + \vec{r}_2 \times \frac{d\vec{r}_1}{dt}$

☐ $\frac{d\vec{r}_2}{dt} \times \vec{r}_1 + \frac{d\vec{r}_1}{dt} \times \vec{r}_2$

Made by: Waqar Siddhu

For More Visit **VU Answer**

For More Visit **VU Answer**

Question No : 36 of 52

Marks: 1 (Budgeted Time 1 Min)

A vector valued function $\vec{F}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$ is continuous if

Answer (Please select your correct option)

VuAnswers.com

☐ Atleast one of its components is continuous.

☐ All of its components are necessarily differentiable.

correct

☐ All of its components are continuous.

☐ Limit exists for all of its components.

Made by: Waqar Siddhu

Question No : 37 of 52

Marks: 1 (Budgeted Time 1 Min)

Given a vector valued function $\vec{F}(t) = \frac{1}{(t-3)}\hat{i} + e^t\hat{j}$ and its anti-derivative $\vec{R}(t) = \ln(t-3)\hat{i} + e^t\hat{j}$, then $\int \vec{F}(t) dt = \dots\dots\dots$

Answer (Please select your correct option)

VuAnswers.com

☐ $\ln(t-3)\hat{i} + e^t\hat{j} + c$

☐ $(t-3)\hat{i} + \frac{e^t}{2}\hat{j} + c$

☐ $(t-3)^{-1}\hat{i} + \frac{e^t}{2}\hat{j} + c$

☐ $\frac{1}{(t-3)}\hat{i} + e^t\hat{j}$

correct

Made by: Waqar Siddhu

Question No : 38 of 52

Marks: 1 (Budgeted Time 1 Min)

For any two vector valued functions $\vec{r}_1(t)$ and $\vec{r}_2(t)$, $\frac{d}{dt}[\vec{r}_1(t) \cdot \vec{r}_2(t)] = \dots\dots\dots$ where " $\times$ " and " $\cdot$ " represent the cross and dot product respectively.

Answer (Please select your correct option)

VuAnswers.com

☐ $\vec{r}_1 \cdot \frac{d\vec{r}_2}{dt} - \frac{d\vec{r}_1}{dt} \cdot \vec{r}_2$

☐ $\vec{r}_1 \cdot \frac{d\vec{r}_2}{dt} + \frac{d\vec{r}_1}{dt} \cdot \vec{r}_2$

☐ $\vec{r}_1 \times \frac{d\vec{r}_2}{dt} + \frac{d\vec{r}_1}{dt} \times \vec{r}_2$

☐ $\vec{r}_1 \cdot \frac{d\vec{r}_2}{dt} - \frac{d\vec{r}_1}{dt} \times \vec{r}_2$

correct

Made by: Waqar Siddhu

For More Visit **VU Answer**

For More Visit **VU Answer**

Question No : 39 of 52

Marks: 1 (Budgeted Time 1 Min)

A single curve can be represented by vector valued function(s).

Answer (Please select your correct option)

VuAnswers.com

- ☐ Two
- ☐ Infinitely many
- ☐ Single
- ☐ Three

correct

Made by: Waqar Siddhu

Question No : 40 of 52

Marks: 1 (Budgeted Time 1 Min)

A function is said to be smooth if it's derivative is on any value of its domain.

Answer (Please select your correct option)

VuAnswers.com

- ☐ continuous and non zero
- ☐ piecewise continuous
- ☐ defined and non zero
- ☐ differentiable

correct

Made by: Waqar Siddhu

Question No : 41 of 52

Marks: 2 (Budgeted Time 4 Min)

Use Wallis sine formula to evaluate $\int_0^{\frac{\pi}{2}} \sin^5 x \, dx$

Answer (Please [click here](#) to Add Answer)

VuAnswers.com

Rich text editor toolbar with icons for Bold, Italic, Underline, Text Color, Background Color, Bulleted List, Numbered List, Indent, Outdent, Link, Unlink, and others. The text area is empty.

Made by: Waqar Siddhu

For More Visit **VU Answer**

For More Visit **VU Answer**

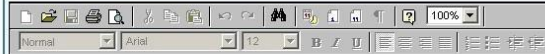
Question No : 42 of 52

Marks: 2 (Budgeted Time 4 Min)

Find Laplace transform of the function $F(t)$ if $F(t) = \cos 3t$.

Answer (Please [click here](#) to Add Answer)

VuAnswers.com



Made by: Waqar Siddhu

Question No : 43 of 52

Marks: 2 (Budgeted Time 4 Min)

State the condition when $\iint_R f(x,y) dA = \int_a^b \int_c^d f(x,y) dx dy = \int_a^b \int_c^d f(x,y) dy dx$ where R is the region of integration.

Answer (Please [click here](#) to Add Answer)

VuAnswers.com



Made by: Waqar Siddhu

Question No : 44 of 52

Marks: 2 (Budgeted Time 4 Min)

Find derivative of the following vector-valued function.

$$\vec{r}(t) = e^t \hat{i} + t^2 \hat{j} + \sec 2t \hat{k}$$

Answer (Please [click here](#) to Add Answer)

VuAnswers.com



Made by: Waqar Siddhu

For More Visit **VU Answer**

For More Visit **VU Answer**

Question No : 45 of 52

Marks: 3 (Budgeted Time 6 Min)

Use Wallis sine formula to evaluate $\int_0^{\frac{\pi}{2}} (\sin^3 x + \sin^4 x) dx$

Answer (Please [click here](#) to Add Answer)

VuAnswers.com

Normal Arial 12 B I U

Made by: Waqar Siddhu

Question No : 46 of 52

Marks: 3 (Budgeted Time 6 Min)

Find Laplace transform of the function $F(t)$ if
 $F(t) = e^{2t} \sin 3t$

Answer (Please [click here](#) to Add Answer)

VuAnswers.com

Normal Arial 12 B I U

Made by: Waqar Siddhu

Question No : 47 of 52

Marks: 3 (Budgeted Time 6 Min)

Find the critical point for the given function $f(x, y) = 6x^2 + xy - 2y^2$ along the line $y = 3x + 1$ at which the absolute extrema of the function can occur.

Answer (Please [click here](#) to Add Answer)

VuAnswers.com

Normal Arial 12 B I U

Made by: Waqar Siddhu

For More Visit **VU Answer**

For More Visit **VU Answer**

Question No : 48 of 52

Marks: 3 (Budgeted Time 6 Min)

What is the arc-length of the curve $\vec{r}(t) = 3\cos t \hat{i} + 3\sin t \hat{j}$ when $0 \leq t \leq 2\pi$?

Answer (Please [click here](#) to Add Answer)

VuAnswers.com

Normal Arial 12 B I U

Made by: Waqar Siddhu

Question No : 49 of 52

Marks: 5 (Budgeted Time 10 Min)

Determine the Fourier co-efficient b_n of the periodic function defined below:
 $f(x) = 2x + 1 \quad 0 < x < 2$

Answer (Please [click here](#) to Add Answer)

VuAnswers.com

Normal Arial 12 B I U

Made by: Waqar Siddhu

Question No : 50 of 52

Marks: 5 (Budgeted Time 10 Min)

Determine whether the following vector field $\vec{F}$ is conservative or not.
 $\vec{F}(x, y, z) = x^2z \hat{i} + y^2x \hat{j} + (y + 2z) \hat{k}$

Answer (Please [click here](#) to Add Answer)

VuAnswers.com

Normal Arial 12 B I U

Made by: Waqar Siddhu

For More Visit **VU Answer**

For More Visit **VU Answer**

Question No : 51 of 52	Marks: 5 (Budgeted Time 10 Min)
If $f(x, y) = (x-4) \ln(xy)$ Find both first order partial derivatives.	
VuAnswers.com	
 100%	
Made by: Waqar Siddhu	
Question No : 52 of 52	Marks: 5 (Budgeted Time 10 Min)
Let $\vec{r}(t) = t^2 \hat{i} + t \hat{j} + (t^2 - 5) \hat{k}$. Find t, such that $\vec{r}(t)$ and $\vec{r}'(t)$ are perpendicular to each other.	
VuAnswers.com	
 100%	
Made by: Waqar Siddhu	

For More Visit **VU Answer**